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Method and apparatus of inspecting printed document

Abstract

An information processing apparatus inspecting printed contents of a printed sheet, the information processing apparatus includes a recognition unit configured to recognize the printed contents printed on the sheet, an acquisition unit configured to acquire an attribute value from information recognized by the recognition unit, a specification unit configured to specify a time required for inspection of the printed contents of the sheet by using the attribute value acquired by the acquisition unit, and a notification unit configured to issue a notification to a user based on a result of comparison between the required time specified by the specification unit and a time limit for inspection of the printed contents.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2012/0182566	12/2011	Sawano	N/A	N/A
2013/0250370	12/2012	Kojima	358/405	H04N 1/00047
2014/0002842	12/2013	Ito	358/1.13	B41F 33/0036
2014/0168709	12/2013	Tokumaru	358/1.18	G06T 7/001
2019/0163335	12/2018	Kamei	N/A	H04N 1/00915

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3435649	12/2018	EP	N/A
2012003335	12/2011	JP	N/A
2017223459	12/2016	JP	N/A
2019084800	12/2018	JP	N/A

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Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a method and an apparatus of inspecting print quality of a printed document.

Description of the Related Art

(2) In recent years, a printing system in which a printed sheet printed by a printing apparatus is inspected by an inspection apparatus during conveyance is known. In the inspection, the inspection apparatus reads an image of the conveyed printed sheet, and determines whether printed contents of the printed sheet are normal, through image analysis of the read image. More specifically, the inspection apparatus can detect missing of a character string or a barcode printed on a printed product, image omission, print defect, page omission, color deviation, printing of an incorrect character string or barcode, and the like. In a case where it is determined that the printed sheet is a defective sheet, the defective sheet is discharged to a sheet discharge destination different from a

sheet discharge destination for a normal sheet. This makes it possible to prevent the defective sheet from being mixed into the normal sheets, which enables an operator to discard the defective sheet. Further, in the above-described system, to improve detection accuracy of the defective sheet of the printed document, a method in which the character string or the barcode to be inspected is read by optical character recognition processing (OCR) or barcode recognition processing, and the inspection is performed through comparison with a prepared answer value is known.

(3) In the printing system requiring the inspection, it is necessary to complete the inspection of the printed sheet while the printed sheet is conveyed and passes through the inspection apparatus. If the inspection is not completed in time, the inspection of a subsequent printed sheet cannot be performed on time. This causes mismatch of an inspection result, and a correct inspection result cannot be obtained. Therefore, it is necessary for the operator of such a printing system to perform inspection setting to reduce an inspection required time, which takes labor.

(4) Japanese Patent Application Laid-Open No. 2012-003335 discusses a technique that calculates priority of an inspection target based on print object information (object type, color, and size) on the inspection target, prior to execution of the inspection, and performs only the inspection executable within a predetermined time limit based on the priority. This makes it possible to efficiently inspect only the inspection target high in priority while lowering difficulty of the inspection setting.

(5) The required time until inspection of each printed sheet is completed is generally varied depending on various conditions such as the number, a data amount, and a size of character strings or barcodes to be inspected. Therefore, in a case where the inspection is performed in light of a read result of the character string or the barcode, the inspection required time is normally varied due to difference of various conditions even among the inspection targets having the same object information. Accordingly, the technique discussed in Japanese Patent Application Laid-Open No. 2012-003335 still has an issue in calculation accuracy of the inspection required time when the inspection is performed in light of the read result of the character string or the barcode. Further, in the technique discussed in Japanese Patent Application Laid-Open No. 2012-003335, the control is performed to inspect only the inspection target high in priority within the time limit. Therefore, inspection of the entire area is not ensured, and it is difficult for the inspection operator to know the actually inspected area.

SUMMARY OF THE INVENTION

(6) The present invention is directed to a printing system that can accurately notify a user of whether a reading inspection of an inspection area of a character string or a barcode preset by the user is performable within a time limit, prior to the inspection.

(7) According to an aspect of the present invention, an information processing apparatus inspecting printed contents of a printed sheet, the information processing apparatus includes a recognition unit configured to recognize the printed contents printed on the sheet, an acquisition unit configured to acquire an attribute value from information recognized by the recognition unit, a specification unit configured to specify a time required for inspection of the printed contents of the sheet by using the attribute value acquired by the acquisition unit, and a notification unit configured to issue a notification to a user based on a result of comparison between the required time specified by the specification unit and a time limit for inspection of the printed contents.

(8) Further features of the present invention will become apparent from the following description of embodiments with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram illustrating configurations of an information processing

apparatus, an inspection apparatus, and an image forming apparatus.

(2) FIG. 2 is a block diagram illustrating the configurations of the information processing apparatus, the inspection apparatus, and the image forming apparatus.

(3) FIG. 3 is a diagram illustrating internal configurations of the information processing apparatus, an inspection unit, the image forming apparatus, and a large-capacity stacker.

(4) FIG. 4 is a flowchart of entire inspection processing.

(5) FIG. 5A is a diagram illustrating an inspection setting screen.

(6) FIG. 5B is a diagram illustrating the inspection setting screen.

(7) FIG. 5C is a diagram illustrating the inspection setting screen.

(8) FIG. 5D is a diagram illustrating the inspection setting screen.

(9) FIG. 6 is a diagram illustrating an inspection area list screen.

(10) FIG. 7 is a flowchart of inspection setting.

(11) FIG. 8 is a flowchart of an inspection time test.

(12) FIGS. 9A and 9B are diagrams each illustrating an alert screen example in a case where an inspection required time exceeds an inspection time limit.

(13) FIG. 10 is a flowchart of the inspection time test.

DESCRIPTION OF THE EMBODIMENTS

(14) Some embodiments of the present invention are described below with reference to drawings. Each of the embodiments of the present invention described below can be implemented solely or as a combination of a plurality of the embodiments or features thereof where necessary or where the combination of elements or features from individual embodiments in a single embodiment is beneficial.

(15) The following embodiments do not limit the invention according to the claims, and all of combinations of characteristics described in the embodiments are not necessarily essential for solving means of the invention.

(16) FIG. 1 is a schematic diagram illustrating configurations of an information processing apparatus **0109**, an inspection apparatus **0108**, and an image forming apparatus **0101** according to a first embodiment. The image forming apparatus **0101** according to the present embodiment is described by using the image forming apparatus **0101** of an electrophotographic system; however, the image forming apparatus **0101** according to the present embodiment may be an image forming apparatus of the other image forming system such as an inkjet system or an offset system.

(17) The image forming apparatus **0101** is connected to the information processing apparatus **0109** through a cable **0112**. The information processing apparatus **0109** is connected to a client computer **0110** and the inspection apparatus **0108** through a network **0113**.

(18) The image forming apparatus **0101** includes a user interface (UI) panel **0102**, a sheet feeding deck **0103**, and a sheet feeding deck **0104**. Further, an optional deck **0105** including sheet feeding decks of three stages is connected to the image forming apparatus **0101**. The image forming apparatus **0101** is, for example, an electrophotographic image forming apparatus. The UI panel **0102** is a user interface including, for example, an electrostatic capacitance touch panel.

(19) The image forming apparatus **0101** further includes an inspection unit **0106** and a large-capacity stacker **0107**. The inspection unit **0106** is connected to the inspection apparatus **0108** through a cable **0114**.

(20) The large-capacity stacker **0107** includes a main tray and a top tray, and several thousand sheets can be loaded on the main tray at a time.

(21) A print job is generated by the client computer **0110**, is transmitted to the information processing apparatus **0109** through the network **0113**, and is managed by the information processing apparatus **0109**. Further, the print job is transmitted from the information processing apparatus **0109** to the image forming apparatus **0101** through the cable **0112**, and the image forming apparatus **0101** performs printing on a sheet. The print job may be generated and managed by the information processing apparatus **0109**, be transmitted to the image forming apparatus **0101**

through the cable **0112**, and be managed by the image forming apparatus **0101**.

(22) The client computer **0110**, the information processing apparatus **0109**, and the inspection apparatus **0108** may communicate with the image forming apparatus **0101** by being connected to the cable **0112**. In other words, the connection mode of the image forming apparatus **0101**, the information processing apparatus **0109**, and the client computer **0110** illustrated in the present embodiment is illustrative, and various connection modes other than the connection mode illustrated in the present embodiment are adoptable as a matter of course.

(23) FIG. 2 is a block diagram illustrating control configurations of the image forming apparatus **0101**, the inspection apparatus **0108**, the large-capacity stacker **0107**, the information processing apparatus **0109**, and the client computer **0110** according to the present embodiment.

(24) A central processing unit (CPU) **0201** performs control and calculation of each unit in the image forming apparatus **0101** through a system bus **0212**. The CPU **0201** executes programs that are stored in a storage unit **0205** and are loaded to a random access memory (RAM) **0202**. The RAM **0202** is a kind of general volatile storage device directly accessible from the CPU **0201**, and is used as a work area for the CPU **0201** or a temporary data storage area. The storage unit **0205** functions as a temporary storage area and a work memory during operation of the image forming apparatus **0101**.

(25) An engine interface (I/F) **0209** performs communication with and control of the printer engine **0210**. A sheet feeding deck I/F **0204** performs communication with and control of a sheet feeding deck **0211**. The sheet feeding deck **0211** is a generic term of hardware configurations of the sheet feeding decks **0103** and **0104** and the optional deck **0105**. A UI panel **0203** is a hardware configuration of the UI panel **0102**, and is a user interface to perform the entire operation of the image forming apparatus **0101**. In the present embodiment, the UI panel **0203** includes an electrostatic capacitance touch panel.

(26) A network interface (hereinafter, NW I/F) **0207** is connected to an NW I/F **0238** of the information processing apparatus **0109** through a cable **0213**, and manages communication between the information processing apparatus **0109** and the image forming apparatus **0101**. In this example, the interfaces connected to the system buses **0212** and **0239** are directly connected to each other; however, the information processing apparatus **0109** and the image forming apparatus **0101** may be connected through, for example, a network, and a connection method is not limited. A video I/F **0206** is connected to a video I/F **0233** through a video cable **0241**, and manages communication of image data between the information processing apparatus **0109** and the image forming apparatus **0101**.

(27) The connection interface of the information processing apparatus **0109** with the image forming apparatus **0101** may have an integrated function of the NW I/F **0238** and the video I/F **0233**. Further, the connection interface of the image forming apparatus **0101** with the information processing apparatus **0109** may have an integrated function of the NW I/F **0207** and the video I/F **0206**.

(28) An accessory I/F **0208** is connected to an accessory I/F **0214** and an accessory I/F **0220** through a cable **0225**. In other words, the image forming apparatus **0101** mutually communicates with the inspection unit **0106** and the large-capacity stacker **0107** through the accessory I/Fs **0208**, **0214**, and **0220**.

(29) A CPU **0216** performs control and calculation of each unit in the inspection unit **0106** through a system bus **0219**, and executes programs that are stored in a storage unit **0247** and are loaded to a RAM **0217**. The RAM **0217** is a kind of general volatile storage device directly accessible from the CPU **0216**, and is used as a work area for the CPU **0216** or a temporary data storage area. The storage unit **0247** functions as a temporary storage area and a work memory during operation of the inspection apparatus **0108**. An inspection apparatus I/F **0215** is connected to an inspection unit I/F **0231** through a cable **0249**. In other words, the inspection unit **0106** communicates with the inspection apparatus **0108** through the inspection apparatus I/F **0215**, and the inspection unit I/F

0231.

(30) An imaging unit **0218** has an imaging function mounted with, for example, a conduct image sensor (hereinafter, CIS), images a sheet passing through the inspection unit **0106**, and transmits a captured image to the inspection apparatus **0108** through the inspection apparatus I/F **0215**. The CIS of the imaging unit **0218** is an example of a sensor, the sensor may be the other type of sensor such as a charge coupled device (CCD) image sensor, and the imaging method is not limited.

(31) A CPU **0221** performs control and calculation of each unit in the large-capacity stacker **0107** through a system bus **0224**, and executes programs that are stored in a storage unit **0248** and are loaded to a RAM **0222**. The RAM **0222** is a kind of general volatile storage device directly accessible from the CPU **0221**, and is used as a work area for the CPU **0221** or a temporary data storage area. The storage unit **0248** functions as a temporary storage area and a work memory during operation of the inspection apparatus **0108**. A sheet discharge unit **0223** performs operation to discharge sheets to the main tray and the top tray, and performs monitoring and control of load states of the main tray and the top tray.

(32) A CPU **0226** performs control and calculation of each unit in the inspection apparatus **0108** through a system bus **0230**, and executes programs that are stored in a storage unit **0228** and are loaded to a RAM **0227**. The RAM **0227** is a kind of general volatile storage device directly accessible from the CPU **0226**, and is used as a work area for the CPU **0226** or a temporary data storage area. The storage unit **0228** functions as a temporary storage area and a work memory during operation of the inspection apparatus **0108**. A page-description language (PDL) analysis unit **0229** reads PDL data such as portable document format (PDF) data, PostScript data, and printer control language (PCL) data, received from the client computer **0110** and the information processing apparatus **0109**, and performs interpretation processing. A display unit **0245** is, for example, a liquid crystal display connected to the inspection apparatus **0108**, and receives user input to the inspection apparatus **0108** and displays a state of the inspection apparatus **0108**.

(33) A CPU **0234** performs control and calculation of each unit in the information processing apparatus **0109** through the system bus **0239**, and executes programs that are stored in a storage unit **0236** and are loaded to a RAM **0235**. The RAM **0235** is a kind of general volatile storage device directly accessible from the CPU **0234**, and is used as a work area for the CPU **0234** or a temporary data storage area. The storage unit **0236** functions as a temporary storage area and a work memory during operation of the information processing apparatus **0109**. A NW I/F **0237** is connected to NW I/Fs **0232** and **0240** through a network. The information processing apparatus **0109** communicates with the inspection apparatus **0108** through the NW I/Fs **0237** and **0232**. Further, the information processing apparatus **0109** communicates with the client computer **0110** through the NW I/Fs **0237** and **0240**.

(34) A CPU **0243** performs control and calculation of each unit in the client computer **0110** through a system bus **0246**, and executes programs that are stored in a storage unit **0244** and are loaded to a RAM **0242**. The RAM **0242** is a kind of general volatile storage device directly accessible from the CPU **0243**, and is used as a work area for the CPU **0243** or a temporary data storage area. The storage unit **0244** functions as a temporary storage area and a work memory during operation of the client computer **0110**.

(35) FIG. 3 is a diagram illustrating internal configurations of the image forming apparatus **0101**, the inspection unit **0106**, and the large-capacity stacker **0107**. The image forming apparatus **0101** receives user input and displays states of printing and the apparatus through the UI panel **0102**. The sheet feeding decks **0103** and **0104** can house various kinds of sheets. Each of the sheet feeding decks **0103** and **0104** can separate only an uppermost sheet of the housed sheets, and convey the sheet to a sheet conveyance path **0305**.

(36) Developing stations **0301** to **0304** form toner images by using colored toners of yellow (Y), magenta (M), cyan (C), and black (K), in order to form a color image. The formed toner images are primarily transferred to an intermediate transfer belt **0306**. The intermediate transfer belt **0306**

rotates in a clockwise direction in FIG. 3, and the toner images are transferred to the sheet conveyed from the sheet conveyance path **0305** at a secondary transfer position **0307**. A fusing unit **0308** includes a pressure roller and a heating roller. When the sheet passes between the rollers, the toner is melted and pressed, and the toner images are accordingly fixed onto the sheet. The sheet having passed through the fusing unit **0308** is conveyed to a sheet conveyance path **0312** through a sheet conveyance path **0309**. In a case where additional melting and pressing are necessary to fix the toner images because of a type of the sheet, the sheet is conveyed to a second fusing unit **0310** through an upper sheet conveyance path after passing through the fusing unit **0308**, is subjected to additional melting and pressing, and is then conveyed to the sheet conveyance path **0312** through a sheet conveyance path **0311**. In a case where an image forming mode is a double-sided image forming mode, the sheet is conveyed to a sheet reversing path **0313**. After the sheet is reversed by the sheet reversing path **0313**, the sheet is conveyed to a double-sided conveyance path **0314**, and image transfer on a second surface is performed at the secondary transfer position **0307**.

(37) In the inspection unit **0106**, CISs **0315** and **0316** are disposed to face each other. The CIS **0315** is a sensor to read an upper surface of the sheet, and the CIS **0316** is a sensor to read a lower surface of the sheet. At a timing when the sheet conveyed to a sheet conveyance path **0317** reaches a predetermined position, the inspection unit **0106** scans the sheet by using the CISs **0315** and **0316**. Scanned images are transmitted to the inspection apparatus **0108** through the inspection apparatus I/F **0215** and the inspection unit I/F **0231**. The CPU **0226** determines whether each of the received images has a defect, and notifies the inspection unit **0106** of a determination result through the inspection unit I/F **0231** and the inspection apparatus I/F **0215** again. The CPU **0216** notifies the large-capacity stacker **0107** of the received determination result through the accessory I/Fs **0214** and **0220**.

(38) The large-capacity stacker **0107** can be loaded with a large number of sheets. The large-capacity stacker **0107** includes a main tray **0324** as a tray loaded with the sheets. The sheet having passed through the inspection unit **0106** enters the large-capacity stacker **0107** through a sheet conveyance path **0319**. The sheet is loaded on the main tray **0324** from the sheet conveyance path **0319** through a sheet conveyance path **0322**. The large-capacity stacker **0107** further includes a top tray **0320** as a sheet discharge tray. The CPU **0221** discharges a sheet on which a defect is detected by the inspection apparatus **0108**, to the top tray **0320**. In a case where the sheet is output to the top tray **0320**, the sheet is conveyed from the sheet conveyance path **0319** to the top tray **0320** through a sheet conveyance path **0321**. A reversing unit **0323** reverses the sheet. The reversing unit **0323** is used to load the sheet on the main tray **0324**. In a case where the sheet is loaded on the main tray **0324**, the sheet is reversed once by the reversing unit **0323** such that a direction of the entering sheet and a direction of the sheet in a loaded state are coincident with each other. In a case where the sheet is conveyed to the top tray **0320**, the sheet is discharged as it is without being flipped at loading. Therefore, reversing operation by the reversing unit **0323** is not performed.

(39) Processing by the system according to the present embodiment is described below with reference to a flowchart and screen examples.

(40) Programs of the image forming apparatus **0101** according to the flow are stored in the storage unit **0205** of the image forming apparatus **0101**, are read out to the RAM **0202**, and are executed by the CPU **0201**. Programs of the inspection apparatus **0108** according to the flow are stored in the storage unit **0228** of the inspection apparatus **0108**, are read out to the RAM **0227**, and are executed by the CPU **0226**. Programs of the information processing apparatus **0109** according to the flow are stored in the storage unit **0236** of the information processing apparatus **0109**, are read out to the RAM **0235**, and are executed by the CPU **0234**. Programs of the client computer **0110** according to the flow are stored in the storage unit **0244** of the client computer **0110**, are read out to the RAM **0242**, and are executed by the CPU **0243**.

(41) FIG. 4 is a flowchart illustrating an entire processing procedure from a reference registration work before start of inspection to execution of the inspection by the inspection apparatus **0108**. The

CPU **0226** of the inspection apparatus **0108** performs the processing.

(42) First, in step **S401**, a user registers a reference image as a reference for the inspection. The inspection apparatus **0108** starts reading of the reference image, and the client computer **0110** executes a print job for registration of the reference image.

(43) Next, in step **S402**, the user performs inspection setting to set various kinds of inspection parameters, an inspection area, and an inspection level. Details thereof are described below.

(44) Next, in step **S403**, the user starts the inspection by the inspection apparatus **0108**, and executes a print job for the inspection by the client computer **0110**. When printing is performed, the inspection unit **0106** detects conveyance of the sheet, and scans the sheet by the CISs **0315** and **0316**. Scanned images are stored in the RAM **0227** of the inspection apparatus **0108**. Each of the stored images is compared with the reference image, and an inspection result is output. The inspection is performed based on set values designated by the user in the inspection setting.

(45) Next, the inspection setting in step **S402** is described in detail with reference to FIGS. **5A** to **5C**, FIG. **6**, and a flowchart of FIG. **7**. The inspection setting is started by selecting the reference image registered in the reference registration.

(46) FIGS. **5A** to **5C** each illustrate an example of an inspection setting screen that is displayed on the UI panel **0102** to receive inspection setting execution operation by the user. In the inspection setting screen, the user arranges inspection areas on a previewed reference image, and individually performs setting on each of the inspection areas.

(47) An area **501** is a preview display area where the reference image is displayed. In a case where a plurality of reference images is provided for a print job including a plurality of pages, the reference image corresponding to each of the pages can be displayed by receiving an operation to switch the reference image to be previewed.

(48) Frames **502** to **504** indicate inspection areas arranged on the previewed image. The inspection areas are arranged by an inspection area arrangement button **506**, and the size and the position of each of the inspection areas can be changed by mouse drag operation by the user. The frame **502** indicates an arrangement example of a picture inspection area.

(49) The frame **503** indicates an arrangement example of a character-string inspection area. The frame **504** indicates an arrangement example of a barcode inspection area.

(50) A button **505** is an inspection area selection button. When the user presses the inspection area selection button **505** and then clicks any of the frames **502** to **504** indicating the inspection areas with a mouse, operation to select the inspection area is received.

(51) A button **506** is an inspection area arrangement button. When the user presses the inspection area arrangement button **506**, selects a type of the inspection area from a displayed pull-down menu, and then selects an area to be designated as the inspection area on the preview display by mouse drag, operation to arrange the inspection area is received.

(52) An item **507** is a setting item to select collation inspection data by a file selection method. The collation inspection data is a reference comma-separated values (CSV) file for data inspection collated when the data inspection is performed. The reference CSV file is a file to be prepared by the user, and is a file in which answer character strings in character string inspection and barcode inspection are listed. At the time of executing the data inspection, reading results of the character-string inspection area and the barcode are collated with the answer character strings listed in the reference CSV file.

(53) An item **508** is a setting item for the selected area, and set values individual for the inspection area selected by the inspection area selection button **505** are displayed.

(54) An item **509** is a setting item for a page range. When the user presses each of buttons, operation to designate a page range to be inspected based on the selected inspection area is received. In a case where nothing is selected, the selected inspection area is disposed only in a page currently displayed in the preview display area **501**. In a case where “same surface as current page” is selected, the selected inspection area is disposed on each of the pages corresponding to the same

surface based on whether the selected inspection area is disposed on a front surface or a rear surface of the sheet. In a case where “all pages” is selected, the selected inspection area is disposed in all of the pages.

(55) An item **510** is a setting item for each type of the inspection area, and the setting item corresponding to the type of the inspection area selected by the inspection area selection button **505** is displayed. FIG. 5A illustrates the setting items for picture inspection displayed when the picture inspection area **502** is selected. The setting items for the picture inspection include an item indicating a defect to be inspected, a detection level of the fine defect, and the like in inspection of a printed document. The inspection item indicates, for example, a circular defect (spot) and a linear defect (streak). The inspection level includes, for example, five stages from level **1** to level **5**, and level **5** can detect a defect thinner and smaller than a defect detectable in the level **1**. Further, the level can be set to each inspection item such that, for example, the spot is detected in the inspection level **5**, and the streak is detected in the inspection level **4**.

(56) FIG. 5B illustrates the setting items for character-string inspection displayed when the character-string inspection area **503** is selected. For each inspection area, an angle of the inspection area, a font of a character string, execution/inexecution of the collation inspection, and a column to be collated in the reference CSV file when the collation inspection is performed are set.

(57) An item to select the font of the character string is displayed, and selection of the font is received from the user.

(58) FIG. 5C illustrates the setting items for barcode inspection displayed when the barcode inspection area **504** is selected. For each inspection area, an angle of the inspection area, a type of barcode, execution/inexecution of the collation inspection, and a column to be collated in the reference CSV file when the collation inspection is performed are set. In the present embodiment, the setting items corresponding to the type of the selected inspection area is displayed to hidden unrelated items; however, all of the items may be displayed while the unrelated items may be disabled, or the unrelated items may be folded and hidden.

(59) A button **511** is an inspection area list display button. When the user presses the button **511**, an inspection area list illustrated in FIG. 6 is displayed.

(60) A button **512** is an inspection setting completion button. When the user presses the button **512**, the inspection setting is completed and stored in the storage unit **0228**, and the inspection setting processing then ends. In a case where the setting has an inconsistency, the inspection setting completion button **512** may be disabled so as to be prevented from being pressed. The inconsistency of the setting indicates that, for example, an inspection area to be subjected to the collation inspection is present, but the setting item **507** for the collation inspection data is not set.

(61) A button **513** is an inspection setting cancel button. When the user presses the button **513**, the setting is discarded and the inspection setting processing ends.

(62) A button **514** is an inspection time test button for the set inspection area. When the user presses the button **514**, it is possible to test whether the inspection for the set inspection area is settled in a time limit. A procedure of processing performed by the inspection apparatus **0108** when the user presses the button **514** according to the present embodiment is described below with reference to drawings.

(63) FIG. 6 illustrates an example of an inspection area list screen that is displayed on the UI panel **0102**, displays a list of the inspection areas under setting, and receives user operation to change the inspection setting. Each row of the list corresponds to any of the arranged inspection areas **502** to **504**, a set value **603** corresponds to the button **509**, and set values **606** to **610** correspond to the item **510**. The user can check and change the set values by selecting each of the inspection areas in the inspection set screen; however, to check the setting of the plurality of inspection areas, a number of operation and a time are required. In contrast, in the inspection area list screen, the user can collectively check and change the set values of the plurality of inspection areas. The inspection area list screen is displayed on the UI panel **0102** in response to pressing of the inspection area list

display button **511** by the user. In the present embodiment, the inspection area list screen is assumed to be displayed on a floating window different from the inspection setting screen; however, the inspection area list screen may be displayed by being transited from the inspection setting screen.

(64) A row **601** is a display row for the set values of each of the inspection areas, and receives selection by mouse click of the user. A context menu may be opened by right click, to receive duplication and deletion of a selected inspection area.

(65) A field **602** is a display field of the type of inspection area. A field **603** is a display field of setting of the page range of the inspection area. The inspection area list screen does not have a preview screen. Therefore, in a case where the page range is changed and the above-described erroneous arrangement occurs, the user cannot visually recognize the erroneous arrangement. Thus, the inspection area list screen does not receive operation to set the page range.

(66) A field **604** is a display field of a sheet where the inspection area is arranged.

(67) A field **605** is a display field of a surface on which the inspection area is arranged.

(68) A field **606** is a display field of angle setting of the inspection area. In the present embodiment, as with the page range, in a case where the angle is changed and erroneous setting occurs, the user cannot visually recognize the erroneous setting. Therefore, the inspection area list screen does not receive operation to set the angle.

(69) A field **607** is a display field of font setting of the inspection area, and receives user operation to change the font.

(70) A field **608** is a display field of setting of the barcode type of the inspection area, and receives user operation to change the barcode type. The barcode type is changed depending on a cell size, a data amount, or the like of the barcode.

(71) A field **609** is a display field of setting whether to perform collation inspection of the inspection area, and receives user operation to change the setting about whether to perform the collation inspection.

(72) A field **610** is a display field of setting of the column in the reference CSV file to be collated with the inspection area, and receives user operation to change the setting of the column.

(73) A button **611** is an inspection area list display end button. In a case where pressing of the button **611** is received, display of the inspection area list ends. An inspection time test button **612** is a button having the function that is the same as the function of the inspection time test button **514**.

(74) FIG. 7 is a flowchart illustrating a processing procedure by the inspection apparatus **0108** when the inspection setting start operation is performed on the inspection apparatus **0108**. The inspection setting screen and the inspection area list screen displayed on the UI panel **0203** of the inspection apparatus **0108** receive the operation, and the CPU **0226** of the inspection apparatus **0108** performs various kinds of processing to perform the inspection setting.

(75) In step **S701**, the inspection apparatus **0108** receives operation to set the collation inspection data from the inspection setting screen. In the present embodiment, the operation to set the collation inspection data is user operation of the collation inspection data setting item **507**.

(76) In step **S702**, the inspection apparatus **0108** receives operation to arrange the inspection area from the inspection setting screen. In the present embodiment, the operation to arrange the inspection area is user operation of the inspection area arrangement button **506**.

(77) In step **S703**, the inspection apparatus **0108** determines whether the setting individual for the inspection area by the user has been performed on the inspection setting screen (FIGS. 5A to 5D) or on the inspection area list screen (FIG. 6). In a case where the setting has been performed on the inspection setting screen (NO in step **S703**), the processing proceeds to step **S704**, and processing in steps **S704** to **S707** are performed. In a case where the setting has been performed on the inspection area list screen (YES in step **S703**), the processing proceeds to step **S708**, and processing in steps **S708** to **S710** are performed.

(78) In step **S704**, the inspection apparatus **0108** receives operation to select the inspection area

from the inspection setting screen (FIGS. 5A to 5D). In the present embodiment, the operation to select the inspection area is user operation of the inspection area selection button **505** displayed on the UI panel **0203**.

(79) In step **S705**, the inspection apparatus **0108** receives operation to set the page range from the inspection setting screen. In the present embodiment, the operation to set the page range is user operation of the page range setting item **509** displayed on the UI panel **0203**.

(80) In step **S706**, the inspection apparatus **0108** receives setting operation for each type of the inspection area, from the inspection setting screen. In the present embodiment, the setting operation for each type of the inspection area is user operation of the setting item **510** for each type of the inspection area displayed on the UI panel **0203**.

(81) In step **S707**, the inspection apparatus **0108** reflects a result of the setting operation performed by the user on the inspection setting screen, in the inspection area list screen (FIG. 6) to synchronize the display contents.

(82) Subsequently, the processing in steps **S708** to **S710** are described. Note that the inspection area list screen (FIG. 6) does not have a preview screen. Therefore, in a case where the page range is changed and the erroneous arrangement of the inspection area occurs, the user cannot visually recognize the erroneous arrangement. Thus, the inspection area list screen does not receive operation to set the page range.

(83) In step **S708**, the inspection apparatus **0108** receives operation to select the inspection area from the inspection area list screen. In the present embodiment, the operation to select the inspection area is user operation of the display row **601** for the set values of each of the inspection areas displayed on the UI panel **0203**.

(84) In step **S709**, the inspection apparatus **0108** receives setting operation for each type of the inspection area, from the inspection area list screen. In the present embodiment, the setting operation for each type of the inspection area is user operation of the display fields **606** to **610** for the respective set values displayed on the UI panel **0203**.

(85) In step **S710**, the inspection apparatus **0108** reflects a result of the setting operation performed by the user on the inspection area list screen, in the inspection setting screen (FIGS. 5A to 5D) to synchronize the display contents.

(86) In step **S711**, the inspection apparatus **0108** determines whether the inspection setting by the user has been completed. More specifically, the inspection apparatus **0108** determines whether the user has pressed the inspection setting completion button **512**. In a case where the inspection setting completion button **512** has been pressed (YES in step **S711**), the processing proceeds to step **S712**. In a case where the inspection setting completion button **512** has not been pressed (NO in step **S711**), the user performs the operation to select the inspection area in step **S703** to continue the inspection setting processing.

(87) In step **S712**, the inspection apparatus **0108** saves the inspection setting in the storage unit **0228**, and then ends the inspection setting processing.

(88) FIG. 8 is a flowchart illustrating a processing procedure by the inspection apparatus **0108** when an inspection time test start instruction is issued to the inspection apparatus **0108**. The CPU **0226** of the inspection apparatus **0108** starts the processing in response to pressing of the inspection time test button **514** displayed on the UI panel **0203** of the inspection apparatus **0108**.

(89) In step **S801**, the inspection apparatus **0108** acquires a sheet size of the print job under the inspection setting.

(90) In step **S802**, the inspection apparatus **0108** calculates an inspection time limit per one page of the inspection job under the inspection setting, from the sheet size acquired in step **S801**.

(91) An example of a method of calculating the inspection time limit in step **S802** is described below. Table A is a table illustrating the inspection time limit of the inspection apparatus **0108** per one page of the inspection job, for the sheet size of the print job.

(92) TABLE-US-00001 TABLE A Time limit per page (Converted based on Sheet size number of

OCR areas) A3 30 B4 30 A4R 30 A4 30 B5R 30 B5 27 A5R 30 11 × 17 30 LGL 30 LTR 30 LTRR
30 STMTR 30 SRA3 30 12 × 18 30 EXEC 28 EXEC-R 30 13 × 19 30 Custom 1 22 Custom 2 24
Custom 3 27

(93) The inspection time limit is represented by the number of areas inspectable per one page of the inspection job while the time required for the inspection of one character-string inspection area is assumed to be 1. For example, in a case where the sheet size of the print job is an A3 size, the inspection time limit corresponds to 30 character-string inspection areas. In a case where the sheet size is a B5 size, the inspection time limit corresponds to 27 areas. The inspection apparatus **0108** holds table information illustrating in the table in the storage unit **0228**. The inspection time limit illustrated in Table A depends on a sheet conveyance speed of the image forming apparatus **0101**. Therefore, it is desirable to hold a table in which, for example, inspection time limits corresponding to the sheet conveyance speeds of various models of the image forming apparatus **0101** are described.

(94) In the present embodiment, as a reference of the inspection time limit, the time required for the inspection of one character-string inspection area is used; however, an actual inspection required time may be measured, and the time limit may be represented by a time representation based on a measurement result, for example, within 500 ms. Alternatively, the time limit may be represented based on other references.

(95) In step **S803**, the inspection apparatus **0108** determines whether, among the inspection areas subjected to the inspection setting described with reference to FIGS. 5A to 5C, FIG. 6, and FIG. 7, an inspection area not yet subjected to calculation of the inspection required time is present. In a case where such an inspection area is present (YES in step **S803**) as a result of the determination, the processing proceeds to step **S804**. In a case where such an inspection area is absent (NO in step **S803**), the processing proceeds to step **S809**. A procedure of calculating the required time is described below.

(96) In step **S804**, the inspection apparatus **0108** acquires setting information about one of the inspection areas subjected to the inspection setting described with reference to FIGS. 5A to 5C, FIG. 6, and FIG. 7.

(97) In step **S805**, the inspection apparatus **0108** performs OCR or barcode recognition processing based on the inspection setting information acquired in step **S804**. At this time, it is determined whether any of the OCR and the barcode recognition processing is performed on the target inspection area, based on the type of inspection area **602**. More specifically, in a case where the type of inspection area **602** is “character-string inspection”, the OCR is performed. In a case where the type of inspection area **602** is “barcode inspection”, the barcode recognition processing is performed.

(98) In step **S806**, the inspection apparatus **0108** acquires attribute values relating to calculation of the inspection required time from a result of the recognition processing performed in step **S805**. In a case of the character string area, examples of the attribute values relating to calculation of the inspection required time include a font type, a font size, and a length of a character string included in the area. In a case of the barcode, examples of the attribute values include a barcode type, a narrowest width, a cell size, the number of cells, an error correction level, and a length of a decoded character string. In the present embodiment, an example in which the inspection required time is calculated by using some of these attribute values is described; however, the inspection required time may be calculated by using all of these attribute values, or may be calculated by using an attribute value other than the above-described attribute values.

(99) In step **S807**, the inspection apparatus **0108** calculates the inspection required time of the inspection area from the attribute values of the inspection area acquired in step **S806**.

(100) Each of Table B and Table C is table information to calculate the inspection required time for each of various kinds of attribute values in the character-string inspection area and the barcode inspection area, acquired by the inspection apparatus **0108** in step **S807**. The table information is

previously held by the storage unit **0228**. As an example, a case where the set inspection area acquired in step **S804** is a barcode inspection area of code **1**, and the attribute values of the area acquired in step **S806** are a cell size of 0.7 mm and a data amount of 800 bytes is considered. In this case, in step **S807**, the inspection apparatus **0108** calculates that the inspection required time of the inspection area is a time corresponding to 10 character-string inspection areas, based on Table C. In the present embodiment, the inspection required time of the character-string inspection area illustrated in Table B is assumed to be a time corresponding to one area irrespective of the attribute values; however, the table information can be set while the inspection required time of the character-string inspection area is assumed to be varied depending on the attribute values. In this case, however, it is necessary to define a combination of reference attribute values, and to define the inspection time limit in Table A, and the inspection required time in Table B and Table C, with the inspection required time in the case of the attribute values as a reference.

(101) TABLE-US-00002 TABLE B Inspection required time (Converted Font based on number type Font size Number of characters of OCR areas) Font 1 Up to 12 pt Up to 30 characters 1 Up to 100 characters 1 Up to 20 pt Up to 30 characters 1 Up to 100 characters 1 Font 2 Up to 12 pt Up to 30 characters 1 Up to 100 characters 1 Up to 20 pt Up to 30 characters 1 Up to 100 characters 1

(102) TABLE-US-00003 TABLE C Inspection required time (Converted Barcode based on number type Cell size Data amount of OCR areas) Code 1 Up to 0.5 mm Up to 100 bytes 1 Up to 1000 bytes 1 Up to 1.0 mm Up to 100 bytes 5 Up to 1000 bytes 10 Code 2 Up to 0.5 mm Up to 100 bytes 2 Up to 1000 bytes 5 Up to 1.0 mm Up to 100 bytes 10 Up to 1000 bytes 15

(103) In step **S808**, the inspection apparatus **0108** adds the inspection required time acquired in step **S807** to a total inspection required time for one page of the inspection job. The processing then proceeds to step **S803** again. In the present embodiment, the inspection apparatus **0108** calculates the total inspection required time; however, the total inspection required time may be calculated by an apparatus different from the inspection apparatus **0108**, and the inspection apparatus **0108** may specify the total inspection required time by acquiring a result of the calculation.

(104) The inspection apparatus **0108** repeats the above-described processing in steps **S803** to **S808** by the number of set inspection areas, to calculate the total inspection required time per one page of the inspection job.

(105) In a case where it is determined in step **S803** that an area not yet subjected to calculation of the required time is absent (NO in step **S803**), the inspection apparatus **0108** determines in step **S809** whether the calculated total inspection required time is greater than the inspection time limit acquired in step **S802**. In a case where the calculated total inspection required time is greater than the inspection time limit (YES in step **S809**) as a result of the determination, the processing proceeds to step **S810**. In a case where the calculated total inspection required time is less than or equal to the inspection time limit (NO in step **S809**), the processing of this flow ends.

(106) In step **S810**, the inspection apparatus **0108** displays a screen prompting the user to revise the inspection setting on the UI panel **0203** because the total inspection required time per one page to the set inspection areas exceeds the time limit for the current inspection job.

(107) FIG. **9A** illustrates an example of the screen displayed in step **S810**. In this example, the above-described screen **901** prompting the user to revise the inspection setting is displayed. Performing such display enables the user to easily know whether it is necessary to correct the inspection setting performed by the user.

(108) FIG. **9B** illustrates another example of the screen displayed in step **S810**. In this example, a screen **902** presenting hint information to reduce the inspection required time together with a message prompting the user to revise the inspection setting as described above is displayed. Such hint information displayed together can be used as a reference when the user performs the inspection resetting again.

(109) Further, the screen displayed in step **S810** may include other contents. For example, among

the inspection areas subjected to the inspection setting, information on the inspection areas may be displayed in descending order of the inspection required time calculated in step **S807**, and information as a reference for determination of the inspection area to be corrected in setting in order to settle the required time within the time limit may be notified together. Such display is within the scope of the present invention as long as information useful for the user to correct the inspection setting is displayed based on the information obtained by recognizing the character string area or the barcode area.

(110) By performing the embodiment in the above-described processing procedure, the user can accurately know whether the inspection required time based on the inspection setting performed by the user is settled within the time limit of the inspection job, prior to execution of the inspection job. This improves usability of the inspection.

(111) In a second embodiment, the fact that it is possible to easily and efficiently determination whether the inspection required time based on the inspection setting performed by the user is settled within the time limit of the inspection job by appropriately performing the present embodiment, is described with reference to drawings. In the description of the present embodiment, descriptions of configurations and processing procedures that are the same as the configurations and the processing procedures of the first embodiment are omitted, and differences are only described.

(112) FIG. **10** is a flowchart illustrating a processing procedure by the inspection apparatus **0108** when the inspection time test start instruction is issued to the inspection apparatus **0108**, according to the present embodiment. The CPU **0226** of the inspection apparatus **0108** starts the processing in response to pressing of the inspection time test button **514** displayed on the UI panel **0203** of the inspection apparatus **0108**.

(113) In step **S802**, the inspection apparatus **0108** acquires the inspection time limit per one page of the inspection job based on time limit table information illustrated in Table D. As illustrated in Table D, the inspection time limit per one page of the inspection job according to the present embodiment is represented in a time format using a unit of millisecond (ms). Note that the time limit table illustrated in Table D is previously held by the storage unit **0228**.

(114) TABLE-US-00004 TABLE D Size Time limit per page (ms)

A3	500	B4	500	A4R	500	A4	500
B5R	500	B5	450	A5R	500	11 × 17	500
LGL	500	LTR	500	LTRR	500	STMTR	500
SRA3	500	12 × 18	500	EXEC	470	EXEC-R	500
13 × 19	500	Custom 1	370	Custom 2	400	Custom 3	450

(115) In step **S802**, the inspection apparatus **0108** acquires the time limit of the inspection job based on the time limit table information described in Table D. Thereafter, the processing proceeds to step **S1001** through the processing in steps **S803** to **S805**.

(116) In step **S1001**, the inspection apparatus **0108** measures a required time when the recognition processing is performed on the target inspection area in step **S805**. The required time may be measured based on time stamp information (not illustrated) in an operation log of the recognition processing in step **S805**, or may be measured by other methods. Alternatively, the required time of the same inspection area may be measured a plurality of times, and the required time may be calculated by calculating a statistically significant value, for example, the maximum value in the measurement results from a plurality of obtained required times, or by calculating the maximum value considering dispersion from distribution of the obtained required times.

(117) In step **S808**, the inspection apparatus **0108** adds the required time measured in step **S1001** in a manner similar to the processing in the first embodiment.

(118) The processing in steps **S803** to **S805**, **S1001**, and **S808** are performed in this order by the number of set inspection areas. In step **S809**, the total inspection required time per one page is compared with the inspection time limit per one page acquired in step **S802**, in a manner similar to that of the first embodiment.

(119) The processing procedure according to the present embodiment described above with reference to FIG. **10** is described by using a specific example of the inspection area.

(120) It is assumed that the inspection setting is performed as illustrated in FIG. 5D. The area **503** is set as the character-string inspection area, the area **504** and two-dimensional codes **515** to **520** are set as the barcode inspection areas, and the inspection time test button **514** is pressed to start the processing described in FIG. 10. A two-dimensional code indicates a QR Code® or a code such as a barcode as illustrated in FIG. 5D. The sheet size of the inspection job is an A4 size, and the time limit per one page of the inspection job calculated in step **S802** based on Table D is 500 ms. Further, the processing in step **S1001** is performed on each of the inspection areas, and the inspection required times of the respective inspection areas and the total value thereof are obtained as illustrated in Table E described below. The column of the inspection area indicates a reference numeral of each of the inspection areas in FIG. 5D.

(121) TABLE-US-00005 TABLE E Inspection area Inspection required time (ms) 503 40 504 15 515 50 516 60 517 50 518 100 519 90 520 110 Total 515

(122) In this case, in the determination in step **S809**, the total inspection required time of 515 ms exceeds the inspection time limit of 500 ms. Therefore, a message indicating that the total inspection required time exceeds the inspection time limit is displayed on the UI panel **0203** in step **S810**.

(123) In the above-described manner, it is possible to perform the present invention by a simpler method without requiring calculation of the inspection required time based on the attribute values obtained from the result of the recognition of the character string or the barcode performed in the first embodiment.

OTHER EMBODIMENTS

(124) In the first embodiment and the second embodiment, as for the processing procedure described with reference to FIG. 8 and FIG. 10, the control to calculate the inspection required time by sequentially performing the processing in steps **S803** to **S808** and **S1001** in series on each of the inspection areas, is described. In a case where the CPU **0226** can perform multithread processing and can perform the processing procedure described in FIG. 8 in parallel, the inspection time limit and the inspection required time can be calculated by performing the processing in steps **S803** to **S808** and **S1001** in parallel in units of the inspection area. For example, a value obtained by multiplying the inspection time limit described in Table A or Table D by the number of threads processable in parallel may be regarded as the inspection time limit, and the value may be compared with the inspection required time calculated in the first embodiment or the second embodiment.

(125) Further, in the case where the image forming mode is the double-sided image forming mode, the CISs **0315** and **0316** of the inspection unit **0106** scan both of front and rear pages at the same time. In this case, the inspection time limit may be set in consideration of whether the inspection is performable in parallel. In other words, in a case where the parallel operation is possible by the multithread, the value obtained by multiplying the inspection time limit described in Table A or Table D by the number of threads processable in parallel may be regarded as the inspection time limit for both surfaces, and the value may be compared with a total inspection required time obtained by adding the inspection required times for both surfaces calculated in the first embodiment or the second embodiment. In a case where both surfaces are inspected by single thread operation, the processing may be performed by regarding a value obtained by dividing the inspection time limit described in Table A or Table D by two, as the inspection time limit per one page.

OTHER EMBODIMENTS

(126) Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific

integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)TM), a flash memory device, a memory card, and the like.

(127) While the present invention has been described with reference to embodiments, it is to be understood that the invention is not limited to the disclosed embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(128) This application claims the benefit of Japanese Patent Application No. 2021-196003, filed Dec. 2, 2021, which is hereby incorporated by reference herein in its entirety.

Claims

1. An information processing apparatus configured to inspect printed contents printed on a sheet, the information processing apparatus comprising: one or more processors; and at least one memory storing executable instructions, which when executed by the one or more processors, cause the information processing apparatus to: perform character recognition processing on a first area on the sheet to recognize printed contents printed included in the first area on the sheet; acquire attribute information related to a character string recognized by performing the character recognition processing on the first area; determine a time required for inspection of the printed contents included in the first area on the sheet based on the attribute information related to the character string; perform code recognition processing on a second area on the sheet to recognize printed contents included in the second area on the sheet, the second area different from the first area; acquire attribute information related to a code recognized by performing the code recognition processing on the second area; determine a time required for inspection of the printed contents included in the second area on the sheet based on the attribute information related to the code; determine a time required for inspection of the sheet based on the time required for inspection of the printed contents included in the first area on the sheet and the time required for inspection of the printed contents included in the second area on the sheet; determine whether the time required for inspection of the sheet exceeds a predetermined time limit for inspection of the sheet; and issue a notification to a user if the time required for inspection of the sheet is determined to exceed the predetermined time limit for inspection of the sheet.
2. The information processing apparatus according to claim 1, wherein the notification indicates that the time required for inspection of the sheet exceeds the predetermined time limit.
3. The information processing apparatus according to claim 1, wherein the predetermined time limit is preset based on a size of the sheet.
4. The information processing apparatus according to claim 1, wherein the notification unit presents the attribute information related to the code to prevent the time required for inspection of the sheet from exceeding the predetermined time limit.
5. The information processing apparatus according to claim 1, wherein the attribute information related to the code includes an attribute value relating to a two-dimensional code and the attribute

information related to the character string includes an attribute value relating to a font printed on the sheet.

6. The information processing apparatus according to claim 5, wherein the two-dimensional code includes a barcode or a quick response (QR) code.

7. The information processing apparatus according to claim 1, further comprising a display, wherein the executable instructions, when executed by the one or more processors, further cause the information processing apparatus to display a setting screen for selection of an inspection area to be arranged on the sheet, the setting screen including an area to set a parameter of the inspection area selected as a target to be arranged on the sheet.

8. The information processing apparatus according to claim 1, wherein the first area is an inspection area designated to handle a font as an inspection target.

9. The information processing apparatus according to claim 1, wherein the second area is an inspection area designated to handle a two-dimensional code as an inspection target.

10. The information processing apparatus according to claim 1, wherein a third area on the sheet is an inspection area designated to handle an image printed on the sheet as an inspection target, the third area being different from the first area and different from the second area.

11. A method of controlling an information processing apparatus configured to inspect printed contents of a printed on a sheet, the method comprising: performing character recognition processing on a first area on the sheet to recognize printed contents included in the first area on the sheet; acquiring attribute information related to a character string recognized by performing the character recognition processing on the first area; determining a time required for inspection of the printed contents included in the first area on the sheet based on the attribute information related to the character string; performing code recognition processing on a second area on the sheet to recognize printed contents included in the second area on the sheet, the second area different from the first area; acquiring attribute information related to a code recognized by performing the code recognition processing on the second area; determining a time required for inspection of the printed contents included in the second area on the sheet based on the attribute information related to the code; determining a time required for inspection of the sheet based on the time required for inspection of the printed contents included in the first area on the sheet and the time required for inspection of the printed contents included in the second area on the sheet; determining whether the time required for inspection of the sheet exceeds a predetermined time limit for inspection of the sheet; and issuing a notification to a user if the time required for inspection of the sheet is determined to exceed the predetermined time limit for inspection of the sheet.

12. A non-transitory computer-readable storage medium storing one or more programs including instructions, which when executed by one or more processors of an information processing apparatus configured to inspect printed contents of a printed on a sheet, cause the information processing apparatus to: perform character recognition processing on a first area on the sheet to recognize printed contents included in the first area on the sheet; acquire attribute information related to a character string recognized by performing the character recognition processing on the first area; determine a time required for inspection of the printed contents included in the first area on the sheet based on the attribute information related to the character string; perform code recognition processing on a second area on the sheet to recognize printed contents included in the second area on the sheet, the second area different from the first area; acquire attribute information related to a code recognized by performing the code recognition processing on the second area; determine a time required for inspection of the printed contents included in the second area on the sheet based on the attribute information related to the code; determine a time required for inspection of the sheet based on the time required for inspection of the printed contents included in the first area on the sheet and the time required for inspection of the printed contents included in the second area on the sheet; determine whether the time required for inspection of the sheet exceeds a predetermined time limit for inspection of the sheet; and issue a notification to a user if

the time required for inspection of the sheet is determined to exceed the predetermined time limit for inspection of the sheet.
